

# SMOKE ZONE

## BUDDY BIN - CIGARETTE WASTE RECYCLING & CIRCULAR ECONOMY PROCESS

Collection | Segregation | Agitation | Nicotine-Containing Liquid Management | Fine-Particle  
Recovery | Material Recycling | Environmental Support

### OUR OBJECTIVE

To create a structured cigarette-waste management system that keeps cigarette butts out of general waste, drains, soil and public areas; separates recoverable fractions; and routes each output through an appropriate recycling, recovery or treatment pathway.

**A responsible waste-management initiative for smoking zones in corporate campuses, IT parks, malls and commercial facilities.**

# 1. Collection & Controlled Handling

## 1. Collection at Source

Dedicated BUDDY BINS are installed at smoking zones so cigarette butts are collected separately from general waste. This supports cleaner premises and reduces the risk of butts entering drains, soil and surrounding landscapes.

## 2. Scheduled Collection

Trained personnel empty the bins at defined intervals using appropriate PPE. Waste is placed in designated containers or bags and transferred to the segregation/processing area.

## 3. Primary Item-Wise Separation

Collected cigarette waste is separated into residual/burnt tobacco, cigarette filters, paper/wrapping material, ash and fine residue, and other non-recoverable contaminants.

| Waste Fraction           | Initial Route                                                |
|--------------------------|--------------------------------------------------------------|
| Residual / burnt tobacco | Separate organic fraction for controlled recovery route      |
| Filter material          | Cleaning, agitation, decontamination and material recovery   |
| Paper / wrapper          | Sorting and paper-recycling route where technically suitable |
| Ash & fine residue       | Separate capture and appropriate treatment/disposal          |
| Other contaminants       | Non-recyclable / controlled waste stream                     |

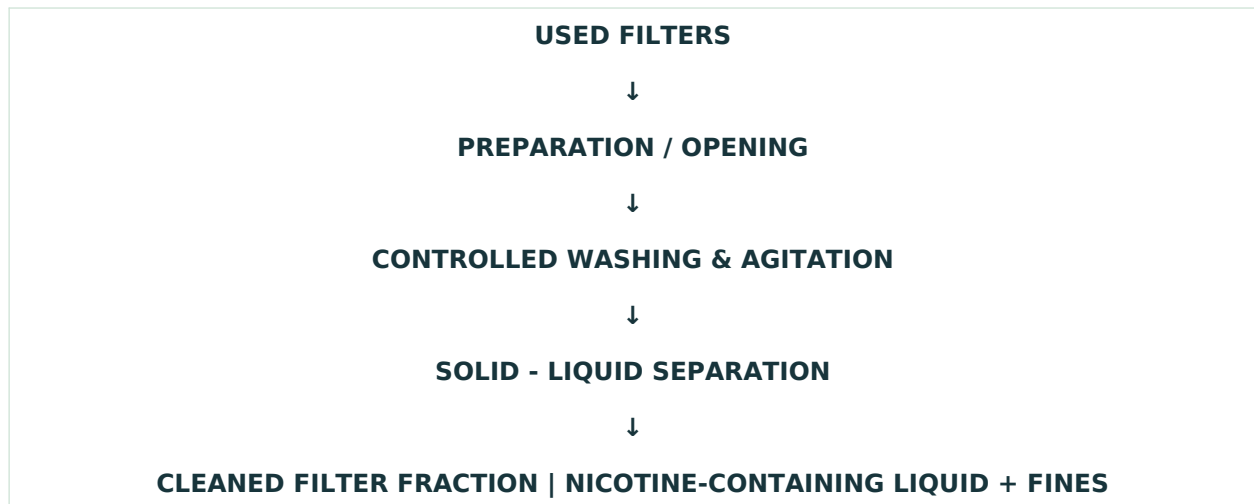
## 2. Filter Agitation, Nicotine & Fine-Particle Management

### 4. Filter Opening / Preparation

Separated filters are prepared for controlled cleaning so the filter material is adequately exposed to the washing medium. Where used, size reduction should be carried out within the defined processing system.

### 5. Controlled Agitation / Washing

Filter material undergoes controlled washing and agitation. The objective is to remove transferable residues from the filter material and move them into a contained liquid stream for subsequent separation and treatment.



### 6. Nicotine-Containing Liquid Separation

During washing, nicotine and other tobacco-related contaminants may transfer into the wash water. This liquid stream is collected separately, stored in designated containers and routed for testing/characterisation and an appropriate authorised treatment, recovery or disposal process.

Important: untreated nicotine-containing wash water should not be described as irrigation water or as a crop-productivity enhancer unless the specific process and end use are supported by laboratory evidence and applicable regulatory approvals.

### 7. Fine & Micron-Sized Residue Management

Agitation can release small suspended particles and fine residues. Screening or filtration is used to capture these solids separately from the liquid stream. 'Micron' refers to particle size; it is not the name of the cigarette-filter material.

**Agitated Liquid → Screening / Filtration → Fine-Particle Capture → Controlled Collection → Appropriate Recovery / Treatment / Disposal**

### 3. Material Recovery & End-Use Pathways

#### 8. Recovered Filter Material

Cigarette filters are primarily made from cellulose acetate fibre. After appropriate cleaning/decontamination, rinsing, dewatering, drying and processing, the recovered material may be evaluated for suitable non-food recycled applications, subject to product-safety, quality and regulatory requirements.

**Cleaned Filter → Rinsing → Dewatering → Drying → Shredding / Processing → Recovered Cellulose Acetate Material → Approved Recycled Applications**

#### 9. Residual Tobacco Fraction

Residual tobacco is segregated from filter and paper fractions. Where technically suitable and legally permitted, the biodegradable fraction may be routed through an authorised organic-waste recovery or composting process. Agricultural use should be claimed only where the finished material has the required testing and approvals.

#### 10. Paper / Wrapper Fraction

Recoverable paper and wrapping fractions are separately sorted and can be directed to an appropriate paper-recycling stream where technically feasible, including sorting, cleaning, pulping/reprocessing and suitable recycled paper products.

#### 11. Ash, Captured Fines & Non-Recyclable Residue

Ash, captured fine particles, contaminated residues and materials that cannot be safely recycled are kept separate from recovered materials and routed through an appropriate authorised treatment or disposal channel.

| Recovered Stream                          | Potential / Appropriate Route                                       |
|-------------------------------------------|---------------------------------------------------------------------|
| Cellulose acetate filter material         | Approved non-food recycled utility / craft / material applications  |
| Paper fraction                            | Suitable recycled paper products                                    |
| Residual tobacco / biodegradable fraction | Authorised organic recovery / composting route where permitted      |
| Nicotine-containing liquid                | Testing, treatment and authorised recovery/disposal                 |
| Fine / micron-sized residue               | Captured separately for appropriate recovery, treatment or disposal |

## 4. How the BUDDY BIN Initiative Supports Nature

### Cleaner Campuses

Dedicated collection helps reduce cigarette-butt litter in smoking zones and surrounding public areas.

### Waste Segregation at Source

Cigarette waste is kept separate from conventional municipal waste, improving traceability and downstream handling.

### Drain, Soil & Landscape Protection

Controlled collection reduces the chance of discarded butts entering drains, soil and landscaped areas.

### Material Recovery

Recoverable filter and paper fractions are separated for suitable recycling pathways instead of treating the entire stream as mixed waste.

### Controlled Contaminant Management

Nicotine-containing liquid, ash and fine residues are isolated from recovered materials and routed for appropriate treatment.

### Circular-Economy Approach

The program focuses on collection, segregation, recovery, recycling and responsible management of residual fractions.

### Environmental Awareness

Buddy Bins and supporting signage encourage responsible disposal behaviour among smoking-zone users.

COLLECT → SEGREGATE → AGITATE / CLEAN → SEPARATE LIQUID & FINES → RECOVER MATERIAL → RECYCLE / TREAT → REDUCE ENVIRONMENTAL IMPACT

### Responsible claims note

Specific claims regarding agricultural fertilizer, nicotine-water use, crop productivity, recycled toys or other consumer products should be supported by the actual process, laboratory testing, product-safety evaluation and applicable regulatory permissions before being presented as established commercial outcomes.